Biology

Topic: Biomolecules

How to Analyse Chemical Composition?

1. Which of the following is correct with respect to the compounds in acid-soluble pool?
1) They are called biomacromolecules.
2) Their molecular weight is around 10,000 Da.
3) They include nucleic acids and lipids.
4) They have molecular weights ranging from 18 to 800 Da.

2. It is said that elemental composition of living organisms and that of inanimate objects (like earth's crust) are similar in the sense that all the major elements are present in both. Then what would be the difference between these two groups? Choose a correct answer from the following.
1) Living organisms have more gold in them than inanimate objects
2) Living organisms have more water in their body than inanimate objects
3) Living organisms have more carbon, oxygen, and hydrogen per unit mass than inanimate objects
4) Living organisms have more calcium in them than inanimate objects.

3. Lipids appear in acid-insoluble fraction during analysis of chemical composition of tissues because
1) They have molecular weight more than 10,000 Da.
2) They are polymeric compounds.
3) On grinding, the bjomembranes are broken into pieces and form insoluble vesicles.
4) They have low molecular weights.

4. Read the given assertion and reason carefully and select the correct option.
Assertion: The acid soluble pool represents roughly the cytoplasmic composition.
Reason: The macromolecules from cytoplasm and organelles become the acid insoluble fraction.
1) If both assertion and reason are true and the reason is the correct explanation of the assertion.
2) If both assertion and reason are true, but reason is not the correct explanation of the assertion.
3) If assertion is true, but reason is false.
4) If both assertion and reason are false.

5. The relative abundance of ____ and _____ is higher in living organisms as compared to earth's crust.
1) Magnesium, silicon

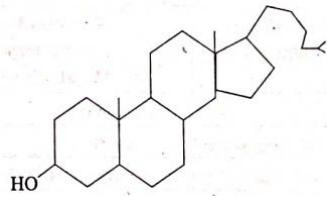
2) Carbon, hydrogen
3) Silicon, oxygen
4) Calcium, magnesium

6. The acid used in the chemical analysis of living tissue is
1) Sulphuric acid
2) Sulphuric acid
3) Hydrochloric acid
4) Trichloroacetic acid

7. A non-polymeric substance in acid-insoluble fraction is
1) Protein
2) DNA
3) Polysaccharide
4) Lipid

8. Identify the wrong statement.
1) A diglyceride has two fatty acids esterified with glycerol.
2) Glycerol is trihydroxy propane.
3) Arachidonic acid has 20 carbon atoms excluding the carboxyl carbon.
4) Lecithin is a phospholipid found is cell membranes.

9. Identify the following structure.



1) Palmitic acid
2) Cholesterol
3) Lecithin
4) Prostaglandin

10. Which of the following statement is correct about lipids?
1) Lipids are small molecular weight compounds and are arranged into structures like cell membrane and other membranes.
2) The membrane fragments in the form of vesicles get separated along with the acid insoluble pool.
3) Lipids remain in the macromolecular fraction.
4) All of the above

Primary and Secondary Metabolites, Biomacromolecules

11. Match the column I with column II and choose the correct option.

Column I		Column II	
A	Toxin	(i)	Vinblastin
B	Drugs	(ii)	Abrin, ricin
C	Polymeric substances	(iii)	Rubber, cellulose

D	Essential oils	(iv)	Lemon grass oil

1) A-(i), B-(ii), C-(iii), D-(iv)
2) A-(ii), B-(i), C-(iv), D-(iii)
3) A-(ii), B-(i), C-(iii), D-(iv)
4) A-(i), B-(iv), C-(ii), D-(iii)

12. Mark the incorrect set with respect to secondary metabolites.
1) Abrin, ricin
2) Rubber, cellulose
3) Monoterpenes, diterpenes
4) Adenosine, sucrose

13. Which of the following secondary metabolites are used as pigments?
(i) Carotenoids (ii) Ricin (iii) Morphine (iv) Anthocyanin
(iii) Morphine (iv) Anthocyanin (v) Abrin (vi) Rubber
1) (i) and (v)
2) (ii) and (vi)
3) (i) and (iv)
4) (ii) and (v)

14. Match the columns I and II.

Column I (Category)		Column II (Secondary Metabolites)	
a	Lectins	i	Codiene
b	Polymeric substances	ii	Abrin
c	Drugs	iii	Concanavalin-A
d	Alkaloids	iv	Vinblastin
e	Toxins	v	Gum

1) a-i, b-ii, c-iii, d-iv, e-v
2) b-i, c-ii, a-iii, e-iv, d-v
3) d-i, e-ii, a-iii, c-iv, b-v
4) b-i, e-ii, a-iii, c-iv, d-v

15. Which secondary metabolite is an alkaloid?
1) Concanavalin A
2) Monoterpenes
3) Morphine
4) Anthocyanin

16. The acid insoluble fraction has
(A) Proteins (B) Nucleic acids (C) Polysaccharides (D) Lipids
1) A and B only
2) A and D only
3) A, B and D
4) A, B, C and D

17. Those chemical compounds which have molecular weights less than one thousand Dalton are usually referred to as
1) Micro molecules

2) Biomolecules
3) Biomacromolecules
4) Both (1) and (2)

18. Which of the following is not strictly macromolecule?
1) Proteins
2) Lipids
3) Polysaccharides
4) Nucleic acids

19. The acid soluble pool represents roughly
1) Cytoplasmic composition
2) Macromolecules from cytoplasm
3) Cell organelles
4) All of the above

20. Refer to the given table:

Component	% of the total cellular mass
Water	70-90
Proteins	(i)
Carbohydrates	3
Lipids	2
Nucleic acids	(ii)

Select the option that correctly identifies (i) and (ii).
1) (i)- 10-15, (ii)-5-7
2) (i)-5, (ii)- 8
3) (i)- 10, (ii)-15
4) (i)-2-4, (ii)-6-10

Proteins, Structure of Proteins

21. Mark the correct statement regarding proteins.
1) Active sites of all enzymes are formed only after acquiring quaternary structure.
2) They are homopolymers of amino acids.
3) GLUT-4 are glucose receptors that inhibit glucose transport into cells.
4) Proteins containing single polypeptide lack quaternary structure.

22. Mark the odd one with respect to fibrous proteins.
1) Collagen
2) Keratin
3) Elastin
4) Enzyme

23. Identify the wrong statement.
1) The most abundant protein in the animal world is collagen..
2) In the primary structure of proteins, monomeric units are joined with each other by peptide bonds.

3) Histamine is formed from histidine.
4) Albumin is the most abundant protein in biosphere.

24. When a long protein chain is folded upon itself like a hollow woollen ball, it gives rise to
1) Primary structure of protein.
2) Secondary structure of protein.
3) Tertiary structure of protein.
4) Quaternary structure of protein.

25. Select the incorrect match.
1) Trypsin-Enzyme
2) Collagen-Intracellular ground substance
3) Insulin-Hormone
4) Antibody-Defence against infections

26. Consider the following statements:
(i) Some proteins act as receptors and help in sensory reception of smell and taste.
(ii) The positional information of amino acids in a protein is provided by its primary structure.
(iii) The first amino acid in a protein is called C-terminal amino acid.
(iv) Primary structure gives us a three-dimensional view of a protein
Which of the above statements are correct?
1) (i) and (ii)
2) (ii) and (iii)
3) (iii) and (iv)
4) (i) and (iv)

27. Which of the following bond plays a major role in stabilising the secondary structure of a protein?
1) Ether bond
2) Peptide bond
3) Hydrogen bond
4) Ester bond

28. Which protein structure does the following represent? Gly -Glu- Ser -Tyr-Cys
1) Primary structure
2) a-Helix
3) b-Pleated
4) Secondary structure

29. Which of the following is an incorrect statement?
1) Cysteine and methionine are sulphur-containing amino acids.
2) In a polypeptide chain, the last amino acid is present at the C-terminal end.
3) Each protein is a polymer of amino acids.
4) Valine is the simplest amino acid.

30. In proteins having more than one polypeptide, the manner in which these individual folded polypeptides are arranged with respect to each other (e.g., linear string of spheres, spheres arranged one upon each other in the form of a cube or plate, etc.) is the architecture of a protein otherwise called the ____ structure

1) Primary
2) Secondary
3) Tertiary
4) Quaternary

31. In quaternary structure of protein, subunits are arranged with respect to each other in the form of

1) Linear string of spheres
2) Spheres arranged in the form of cube
3) Spheres arranged in the form of plate
4) All ofthe above

32. Of course, only some portions of the protein thread are arranged in the form of a helix. Other regions of the protein thread are folded into other forms in what is called the

1) Secondary structure
2) Quaternary structure
3) Tertiary structure
4) Primary structure

Polysaccharides

33. Identify the wrong statement.

1) Lactose is also called milk sugar.
2) Hydrolysis of maltose yields two ß-glucose molecules.
3) The functional group in glucose is aldehyde, while fructose is ketone.
4) Ribose and deoxyribose are found in nucleic acids.

34. Which of the following polysaccharides are made up of glucose monomers?

1) Starch, glycogen, cellulose
2) Chitin, cellulose, peptidoglycan
3) Inulin, chitin, starch
4) Starch, inulin, lactose

35. Identify the heteropolysaccharide among the following sugars:

1) Cellulose
2) Inulin
3) Starch
4) Peptidoglycan

36. Identify the wrong statement.

1) In a polysaccharide, the right end is called reducing end and left end is called non-reducing end
2) Maltose is commonly called malt sugar and found in germinating seeds.
3) Fructose is called fruit sugar.
4) Inulin is a homopolymer of glucose.

37. Chitin, the second most abundant natural polysaccharide, is a linear polymer composed of repeating units called
1) Sucrose
2) N-acetylglucosamine
3) Cellulose
4) Inulin

38. Identify the wrong statement with respect to cellulose.
1) It is the most abundant organic compound on earth.
2) It is found in the cell wall of plants.
3) It has β-1,4 linkage between the glucose molecules.
4) It can hold I_2 molecules in the helical portion.

39. Starch and glycogen differ in
1) Type of monomers in them.
2) Type of cells in which they are present as stored food.
3) Type of glycosidic bond between glucose molecules in chain.
4) None of the above

40. Read the given statements and select the correct option.
(i) Fructose is the sweetest sugar.
(ii) The right end of a polysaccharide chain is the reducing end, while the left end is the non-reducing end.
(iii) Cellulose can hold iodine molecules in its helical portion and gives blue colour with it.
(iv) Plant cell walls are made of chitin.
1) Statements (ii) and (iii) are correct.
2) Statements (i) and (iv) are false.
3) Statements (i) and (ii) are correct.
4) All statements are correct.

41. Identify the wrong statement with respect to chitin.
1) It is a homopolysaccharide of N-acetylglucosamine.
2) It is the second most abundant biopolymer on earth.
3) It is a structural polysaccharide.
4) It is found in the cell wall of plants.

42. Read the following statements:
(i) Paper made from plant pulp and cotton fibre is cellulosic.
(ii) Ribose differs from deoxyribose in having hydroxyl group at the second carbon.
(iii) Inulin is a polymer of sucrose

(iv) Exoskeletons of arthropods have a complex polysaccharide called chitin.

Which of the above statements are correct?

1) (i) and (iii)
2) (iii) and (iv)
3) (ii) and (iii)
4) (i), (ii), and (iv)

43. Which of these is a storehouse of energy in plant tissues?

1) Starch
2) Cellulose
3) Glycogen
4) Glycogen

44. Which of these is not true about cellulose?

1) It is the second most abundant organic polymer
2) It has linear chains
3) It is made of glucose subunits
4) It is a homopolymer

Nucleic Acids

45. Which of the following is not formed by the hydrolysis of a nucleoside?

1) Sugar
2) Phosphoric acid
3) Nitrogenous base
4) Both (1) and (3)

46. Identify the wrong statement with respect to the structure of DNA?

1) One turn of the helical structure of DNA has 10 base pairs.
2) The nitrogen bases are projected perpendicular to the sugar-phosphate backbone but face inside.
3) A and G of one strand always base pair with C and T, respectively on the other strand.
4) There are two hydrogen bonds between A and T, and three hydrogen bonds between G and C.

47. In a nucleic acid, a phosphate moiety links the

1) 3'-carbon of the sugar of one nucleotide to the 5'-carbon of the sugar of the preceding nucleotide.
2) 3'-carbon of the sugar of one nucleotide to the 3'-carbon of the sugar on the other strand.
3) Sugar of one strand with the nitrogen base on the other strand.
4) 3'-carbon of the sugar of one nucleotide to the 5'-carbon of the sugar of the succeeding nucleotide.

48. Which of the following is an incorrect statement?

1) Adenine and guanine are substituted purines.

2) Nucleic acids are formed by the polymerisation of a large number of nucleotides.
3) The backbone of DNA is built up of alternate sugar and phosphate groups.
4) The two strands of DNA are parallel.

49. In B-DNA, the pitch would be
1) 3.4 Å
2) 0.34 Å
3) 3.4 nm
4) 0.34 nm

50. _____ bond is present between a nitrogen base and pentose sugar in a nucleoside.
1) Phosphodicster
2) Glycosidic
3) Peptide
4) None of the above

51. Which of the following is an incorrect statement?
1) Adenine is found in both DNA and RNA.
2) The pitch of B-DNA is 0.34 nm.
3) The two strands of DNA are complimentary to each other
4) The basic unit of nucleic acid is nucleotide.

52. Adenine, guanine and thymine are all
1) Nitrogenous bases
2) Nucleosides
3) Purines
4) Pyrimidines

53. The heterocyclic compounds in nucleic acids are the
1) Nucleotides
2) Nitrogenous bases
3) Nucleosides
4) None of the above

54. Which of the following is/are component(s) of nucleotides?
1) A monosaccharide
2) A heterocyclic compound
3) A phosphoric acid or phosphate
4) All of the above

55. The bond between phosphate and hydroxyl group of a sugar is
1) Glycosidic bond
2) Ester bond
3) Phosphodiester bond
4) Hydrogen bond

56. Which of the following nucleotide sequence contains six purine bases?
1) UGACUAAAGGC

2) TTAAGGCGAAG
3) UUUCCGCCGAU
4) TATAGCGCGGC

Nature of Bond Linking Monomers In a Polymer

57. Refer to the given statement and select the option that correctly fills blanks in it.
In a polypeptide or a protein, amino acids are linked by a _(i)_bond which is formed when the _ (ii) _ group of one amino acid reacts with the_(iii)_ group of the next amino acid with the elimination of a_(iv)___

	(i)	(ii)	(iii)	(iv)
1)	Phosphodiester	amino	carboxyl	Carbon dioxide
2)	Peptide	carboxyl	amino	water moiety
3)	Ester	carboxy	amino	oxygen
4)	Glycosidic	carboxyl	carboxyl	water moiety

58. Refer to the given statement and select the option that correctly fills blanks in it.
In a polysaccharide the individual monosaccharides are linked by a ____ bond. This bond is formed by ____
1) Glycosidic, dehydration
2) Peptide, carboxylation
3) Phosphodiester, dehydration
4) Ester, isomerisation

59. The bond between the phosphate and hydroxyl group of sugar is called
1) Peptide bond
2) Ester bond
3) Glycosidic bond
4) Jonic bond

60. Select the correct option regarding Watson and Crick model of DNA.
(i) DNA exists as a double helix.
(ii) The two strands of polynucleotides are antiparallel i.e., run in the opposite direction.
(iii) The backbone is formed by the sugar phosphate-sugar chain.
(iv) The nitrogen bases are projected more or less parallel to this backbone but face outside.
1) Only (i) and (ii) are correct
2) (i), (ii) and (iv) are correct
3) (ii), (iii) and (iv) are correct
4) (1), (ii) and (iii) are correct

61. There are ___ hydrogen bonds between A and T and ___ hydrogen bonds between G and C.
1) 2,3
2) 2,4
3) 4,3

4) 3,2

Enzymes

62. Select the option which is incorrect with respect to enzymes.
1) The class of enzymes contained in lysosome is hydrolases.
2) Addition of a large amount of substrate will reverse inhibition in case of competitive inhibitor.
3) Cyanide inhibits cytochrome oxidase by noncompetitive mechanism.
4) The effect of non-competitive inhibitor, for example cyanide, can be overcome by increasing the concentration of substrate.

63. Select the incorrect option among the following:
1) All coenzymes are cofactors, but all cofactors are not coenzymes.
2) All enzymes, without exception, use cofactors for their activity.
3) Enzymes get damaged at high temperature
4) Cofactors are non-protein constituents bound to the enzyme to make the enzyme catalytically active.

64. During conversion of substrate to product, transition states formed are
1) Stable and transient
2) Temporary and unstable
3) Permanent altered states
4) None of the above

65. Which of the following statement (s) is/are correct with respect to the given reaction?

$$CO_2 + H_2O \rightleftharpoons H_2CO_2$$

1) In the absence of enzyme, 200 molecules of carbonic acid are formed per hour.
2) In the presence of carbonic anhydrase, 600,000 molecules of H_2CO_3 are formed per second.
3) The presence of enzyme increases the rate of reaction 10 million times.
4) All are correct.

66. Read the following statements:
(i) An active site of an enzyme is a crevice or pocket into which substrate fits.
(ii) Enzymes work efficiently at high temperature and high pressures, while inorganic catalysts get damaged at high temperatures.
(iii) Nucleic acids that behave like enzymes are called ribozymes.
(iv) Uncatalysed reactions proceed at rates vastly higher than that of catalysed ones.

Which of these statements are incorrect?
1) (i) and (ii)
2) (ii) and (iii)
3) (iii) and (iv)
4) (ii) and (iv)

67. Enzymes are biocatalysts. They catalyse biochemical reactions. In general, they reduce activation energy of reactions. Many physico-chemical processes are enzyme mediated. Some examples of enzyme mediated reactions are given below. Tick the wrong entry.

1) Dissolving CO_2 in water
2) Unwinding the two strands of DNA
3) Hydrolysis of sucrose
4) Formation of peptide bond

68. Enzyme catalysts differ from inorganic catalysts as

1) Enzyme catalysts work efficiently at high temperatures and high pressures; inorganic catalysts get damaged at high temperatures (say above 40°C)
2) Inorganic catalysts work efficiently at high temperatures and high pressures, while enzymes get damaged at high temperatures (say above 80–90°C)
3) Inorganic catalysts work efficiently at high temperatures and high pressures, while enzymes get damaged at high temperatures (say above 40°C)
4) Only inorganic catalyst decreases activation energy not enzyme catalysts.

69. _____ is the cofactor for the enzyme carboxypeptidase.

1) Calcium
2) Zinc
3) Magnesium
4) Iron

70. Read the following statements and choose the correct option.

1) Thermal stability is an important quality of enzymes isolated from all living organisms.
2) An enzyme, like any other protein has a primary, secondary and tertiary structures.
3) Active site of an enzyme is a crevice or pocket into which the substrate and product fits.
4) All of the above

Answer Key

1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**
4	3	3	2	2	4	4	3	2	4	3	4	3	3	3
16	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
4	4	2	1	1	4	4	4	3	2	1	3	1	4	4
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**	**41**	**42**	**43**	**44**	**45**
4	1	2	1	4	4	2	4	2	3	4	4	1	1	2
46	**47**	**48**	**49**	**50**	**51**	**52**	**53**	**54**	**55**	**56**	**57**	**58**	**59**	**60**
3	4	4	3	2	2	1	2	4	2	4	2	1	2	4
61	**62**	**63**	**64**	**65**	**66**	**67**	**68**	**69**	**70**					
1	4	2	2	4	4	1	3	2	2					